EEML 2026 Extended Abstract

Optimizing Multi-Agent Orchestration and Inference Latency in Real-Time Sports Intelligence Systems

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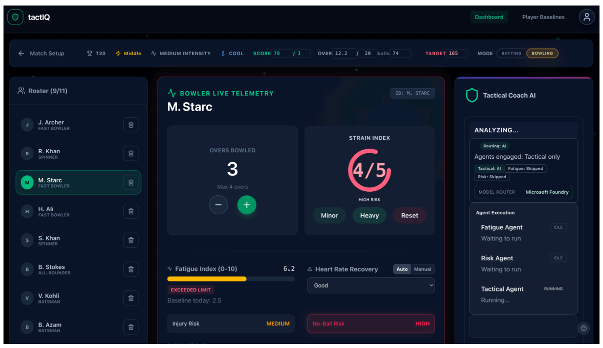

Figure 1: *The TaqtIQ Platform Interface.Live telemetry dashboard displaying real-time strain indices and the model routing layer*

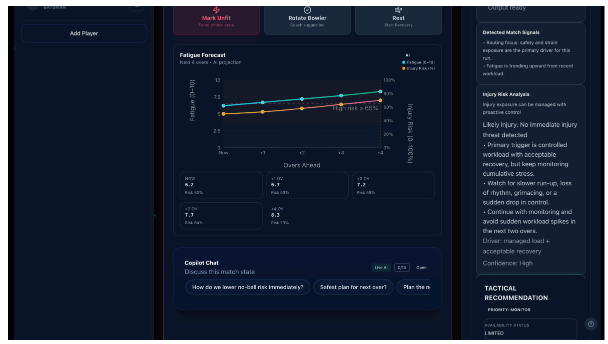

Figure 2: *Deep-dive fatigue forecasting and tactical recommendation engine, showcasing the agent-derived actionable insights and risk projections.*

Abstract— This paper presents tactIQ, a multi-agent AI framework for real-time sports decision support. Traditional monolithic LLM systems struggle with latency and contextual noise when processing simultaneous telemetry signals such as player workload and match state. tactIQ addresses this by using a Router–Worker architecture that decomposes reasoning into specialized agents for fatigue monitoring, injury risk analysis, and tactical strategy. The system is deployed on Microsoft Azure and dynamically routes AI reasoning through the Microsoft Foundry Model Router, enabling scalable and explainable match decision support.

Details for taqtIQ are available at: https://github.com/MahroshAtif2005/tactIQ

1. Introduction

Real-time sports analytics requires rapid interpretation of workload signals, match pressure, and tactical context. While modern tracking systems collect large volumes of player performance data, coaches must still manually interpret this information during live matches. Large language models provide strong reasoning capabilities but struggle with simultaneous telemetry streams and contextual noise. tactIQ addresses this challenge by introducing a multi-agent system that distributes analytical tasks across specialized agents coordinated through a Router–Worker orchestration layer.

2. System Architecture

tactIQ is deployed as a cloud-native system on Microsoft Azure. The architecture follows a Router–Worker pattern in which an orchestration layer receives structured match context and dynamically assigns tasks to specialized agents. These agents focus on fatigue monitoring, injury risk assessment, and tactical reasoning. This modular design allows new analytical agents to be integrated without modifying the core orchestration pipeline.

3. Methodology: Multi-Agent Orchestration

tactIQ converts match context and player workload data into structured signals processed by the orchestration layer. Azure Functions build a unified decision context containing match phase, player role, fatigue indicators, and workload trends. Each agent submits reasoning requests through the Microsoft Foundry Model Router, which dynamically selects the most appropriate language model for the request. The outputs of the agents are then combined into a final tactical recommendation returned to the coaching interface.

4. Results and Evaluation

Simulated match telemetry scenarios were used to evaluate system responsiveness and reasoning stability. The multi-agent architecture reduced analytical noise compared to monolithic LLM prompts by isolating decision dimensions across specialized agents. The system maintained stable response times during repeated orchestration cycles and demonstrated improved clarity in tactical recommendations under simulated match pressure scenarios.

5. Conclusion and Future Work

This work demonstrates that multi-agent orchestration provides an effective framework for real-time sports intelligence systems. By distributing reasoning across specialized agents, tactIQ improves analytical clarity and scalability compared to monolithic LLM pipelines. Future work will extend the platform with multimodal analysis, enabling agents to incorporate computer vision signals from live match footage.

References

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4. Atif, M. (2026). *tactIQ: Multi-Agent Orchestration for Sports Intelligence*. GitHub Repository. https://github.com/MahroshAtif2005/tactIQ